

Interacting Multipoles of Rank 1 and 3:

$$T_{\alpha\beta\gamma\delta}\mu_{\alpha}^A\Omega_{\beta\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\alpha, \delta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\alpha, \gamma) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \end{aligned} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta}\mu_{\alpha}^A\Omega_{\beta\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\gamma\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Omega_{\beta\beta\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Omega_{\alpha\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Omega_{\alpha\beta\delta}^B \\ &-15 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Omega_{\alpha\gamma\delta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Omega_{\alpha\beta\beta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Omega_{\alpha\beta\beta}^B \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Omega_{\alpha\beta\beta}^B \end{aligned} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^A \cdot \Omega_{\beta\gamma\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\beta\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \mu_\alpha^A \cdot \Omega_{\beta\beta\delta}^B \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\delta}^B \\ &-15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Omega_{\alpha\gamma\delta}^B \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \end{aligned} \right] \quad (2b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Omega_{\beta\gamma\delta}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^A \cdot \Omega_{\beta\gamma\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\beta\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\beta\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\gamma}^B \\ &-15 \cdot R_z^2 \cdot R_\gamma \cdot R_\beta \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\gamma}^B \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \end{aligned} \right] \quad (2c)$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Omega_{\beta\gamma\delta}^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^A \cdot \Omega_{\beta\gamma\gamma}^B \\ -30 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\beta\beta\gamma}^B \\ -45 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\gamma}^B \\ +9 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \end{bmatrix} \quad (2d)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta} \mu_\alpha^A \Omega_{\beta\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Omega_{zzzz}^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Omega_{z\gamma\gamma}^B \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Omega_{z\beta\beta}^B \\ -45 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Omega_{zz\alpha\alpha}^B \\ +9 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \end{bmatrix} \quad (3a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{zz\alpha\alpha}^B \\ +9 \cdot R_z^4 \cdot \mu_\alpha^A \cdot \Omega_{\alpha\beta\beta}^B \end{bmatrix} \quad (3b)$$

Simplify Dipoles:

$$T_{\alpha\beta\gamma\delta} \mu_\alpha^A \Omega_{\beta\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{yzz}^B \\ +9 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{y\beta\beta}^B \end{bmatrix} \quad (4a)$$

$$\xRightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{yzz}^B \\ +9 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{y\alpha\alpha}^B \end{bmatrix} \quad (4b)$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta} \mu_\alpha^A \Omega_{\beta\gamma\delta}^B = R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{yzz}^B \\ +9 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{xxy}^B \\ +9 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{yyy}^B \\ +9 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{yzz}^B \end{bmatrix} \quad (5a)$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -36 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{yzz}^B \\ +9 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{xxy}^B \\ +9 \cdot R_z^4 \cdot \mu_y^A \cdot \Omega_{yyy}^B \end{bmatrix} \quad (5b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta} \mu_\alpha^A \Omega_{\beta\gamma\delta}^B = \begin{bmatrix} -\frac{12}{5} \cdot R_z^{-5} \cdot \mu_y^A \cdot \Omega_{yzz}^B \\ +\frac{3}{5} \cdot R_z^{-5} \cdot \mu_y^A \cdot \Omega_{xxy}^B \\ +\frac{3}{5} \cdot R_z^{-5} \cdot \mu_y^A \cdot \Omega_{yyy}^B \end{bmatrix} \quad (6)$$